ShiftSense — Project and User Guide

EXL × IIM Lucknow Live Action Project · Group 24 (Saswata Saha, Supratik Sarkar) · October 2026

> All data in ShiftSense is synthetic (seeded generator). Figures labelled *Simulated* or *Modelled estimate* are not measured results.

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1. What ShiftSense is

ShiftSense is a workforce-operations **decision layer** for SLA-driven BPO contact centres. It continuously compares demand with the skilled capacity that is available, detects coverage risk, and recommends the least-disruptive feasible action to restore service within cost and policy limits. Around that loop it runs the people processes a contact centre needs: leave, swaps, attendance, HR records and governance.

The question the PoC answers: can ShiftSense create a better workforce plan, and restore coverage faster and at lower incremental cost than a credible manual process, without breaking hard workforce rules?

Loop: Predict → Plan → Execute → Detect → Recover → Learn.

Step	What happens
Predict	Seasonal forecast of calls per queue and 30-minute slot, with an uncertainty band (no ML)
Plan	Erlang C converts the forecast into required people; a CP-SAT optimiser builds the roster under every hard rule
Execute	The Analyst validates and publishes the roster; agents see their shifts and check in
Detect	Absences, no-shows and volume spikes open a gap per queue; risk is scored (Critical / At risk / Healthy)
Recover	Candidate fixes are generated, filtered on hard rules, simulated and ranked; a team lead approves, modifies or rejects
Learn	Every decision and reason is logged; KPIs and the evidence table compare approaches

2. Results on the synthetic data (simulated)

Hypothesis	Result	Measured (simulated)
H1 Planning effort falls ≥ 70 %	**Pass**	92 % less effort (400 → 30 min, modelled); coverage 90.9 % vs 75.4 %
H2 Under-covered workload falls ≥ 20 % at equal or lower paid hours	**Pass**	24.6 % → 9.1 % at the same 3,824 paid hours

Hypothesis	Result	Measured (simulated)
H3 Median Time-to-Cover < 10 min	**Pass**	7.0 min (modelled) vs 58 min manual on the same disruptions
H4 Equal coverage restored more cheaply than overtime-first	**Pass**	median ₹4,950 vs ₹9,500, cheaper in 8 of 8 scenarios (simulated rates)
H5 Zero hard-rule violations	**Pass**	0 across 3 plans, 1 re-plan and 32 recoveries (independent checker)

Full ablation: `evidence/test-results/workflow-validation/ablation-simulated.csv`.

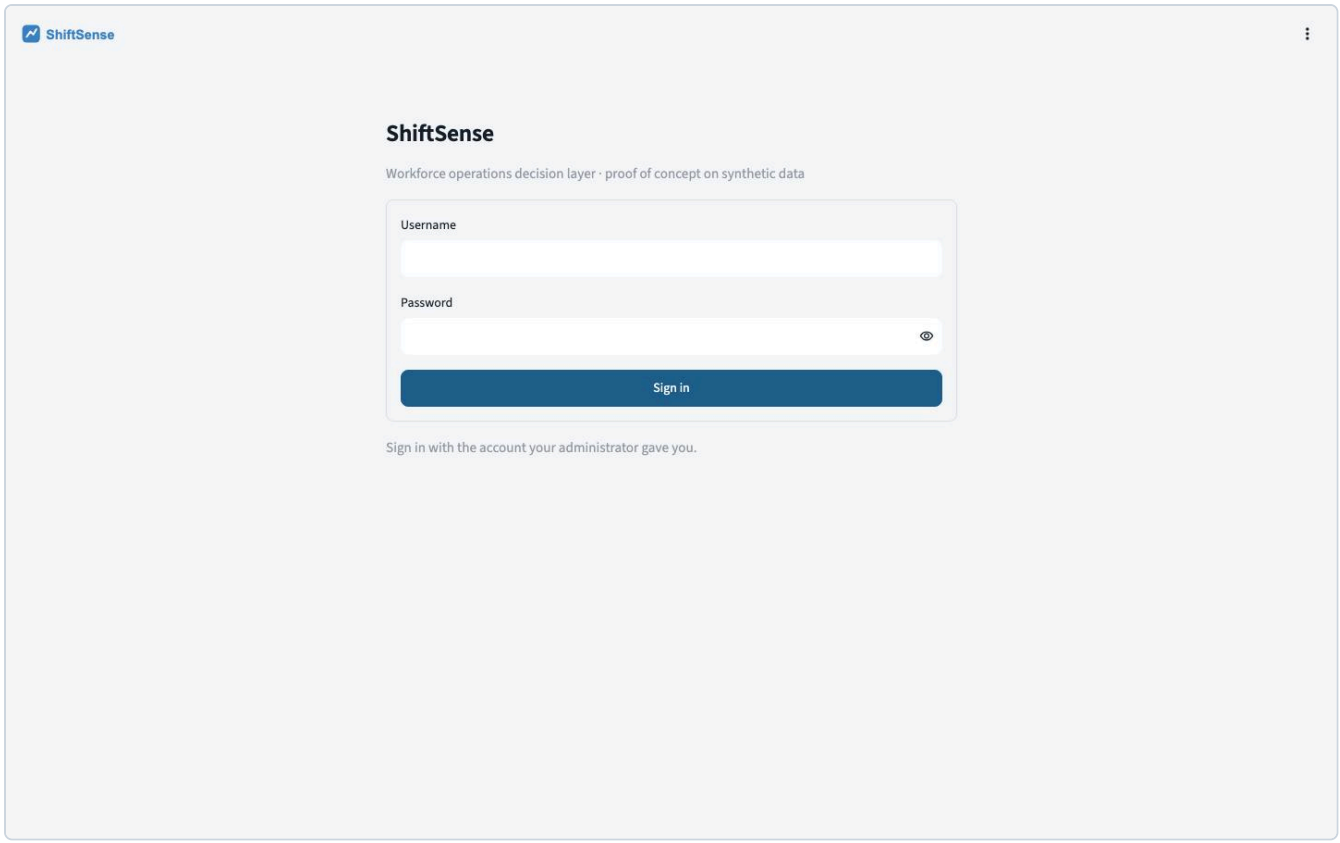
3. Roles

Six roles. Every permission is checked on the server for every action, and nobody can approve their own request.

Role	Purpose	Main screens
Agent	See shifts, check in, ask for leave or swaps, report a problem	Home · My shifts · Inbox & requests · My history
Team Lead	Run their own team: attendance, requests, staffing alerts	Overview · Attendance · Team schedule · Requests · Alerts & decisions · Team information
Ops	Escalations, coverage and cost decisions, carried risk	Overview · Coverage & risk · Alerts & escalations · Recovery · Cost & staffing · Follow-ups, data & history
Analyst	Forecast, plan, publish, data, demo scenario	Planning (Forecast & staffing, Roster optimization, What-if, Review & publish, Benchmarks & history) · Data (overview, add/update, changes, situations, downloads) · Demo (scenario, clocks)
HR	Employees, contracts, leave, balances, attendance corrections, HR cases, HR policy	People overview · Employees · Leave & balances · Attendance · Reviews & approvals · Restricted cases · HR policies · Reports & exports
Admin	Oversight only: accounts, governance, audit, system status (no operational decisions)	Overview · Accounts & access · Governance & decisions · Audit & integrity · System status

4. Getting started

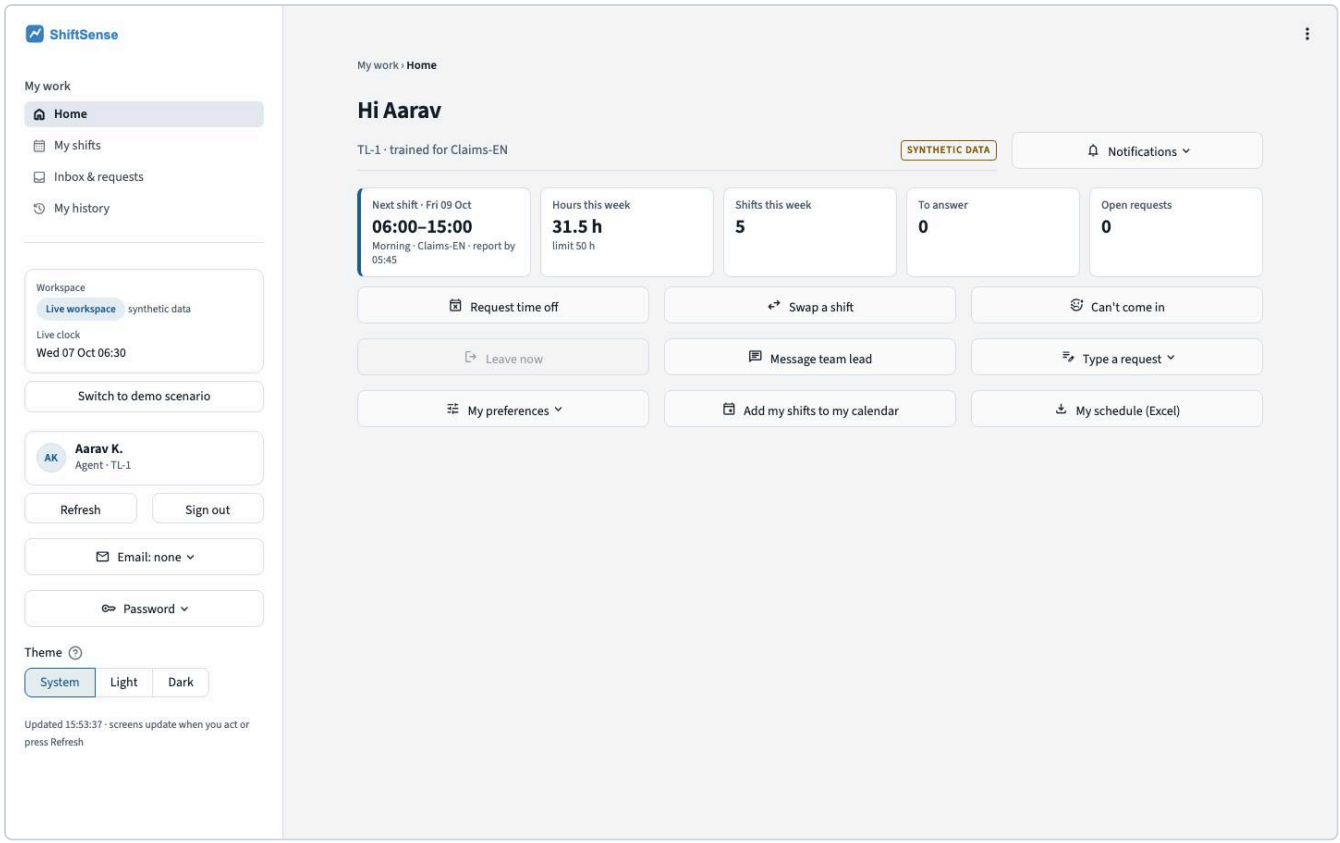
1. Open the address you were given and sign in (username and password, or **Sign in with your company account** when single sign-on is configured).
2. The left sidebar shows your role's screens, the workspace (Live, or the clearly marked Demo scenario), your name, **Refresh**, **Sign out**, email settings, **Password** and **Theme** (System / Light / Dark).
3. Screens do not refresh by themselves: press **Refresh** to see changes made by others.
4. Every screen carries a breadcrumb, a **Synthetic data** badge and a **Notifications** menu.



5. Using ShiftSense by role

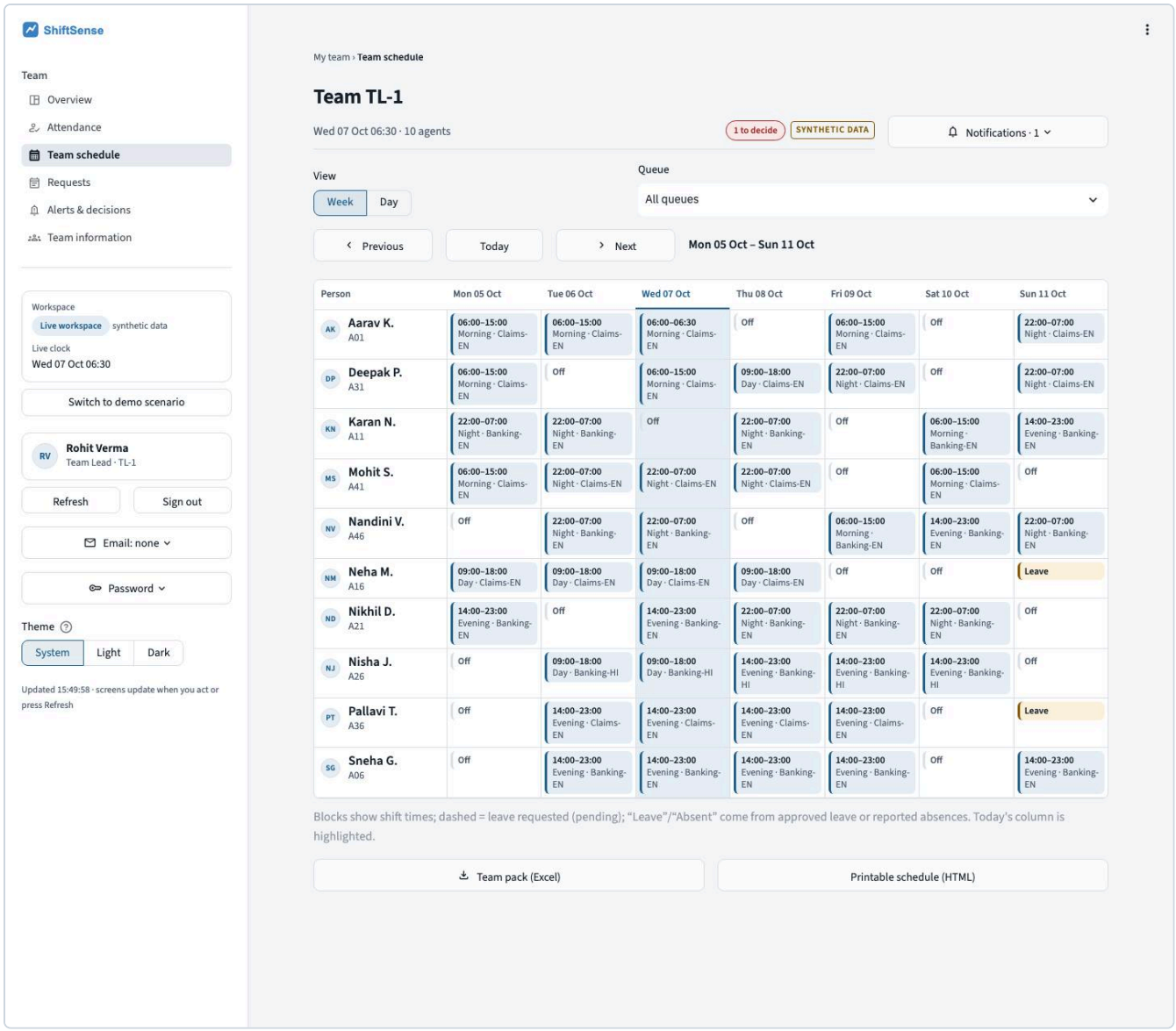
5.1 Agent

- **Home:** today's shift, the check-in button when the window is open (30 min before to 1 h after the start), and anything waiting for you.
- **My shifts:** a week calendar of your shifts with their record status; a timeline view is available.
- **Inbox & requests:** ask for time off (form or plain text such as "sick leave on Friday"), ask a trained colleague to swap, reply to cover offers, withdraw a pending request. A Red coverage result offers alternatives.
- **My history:** everything recorded about your shifts; *Report a record problem* sends a correction request.



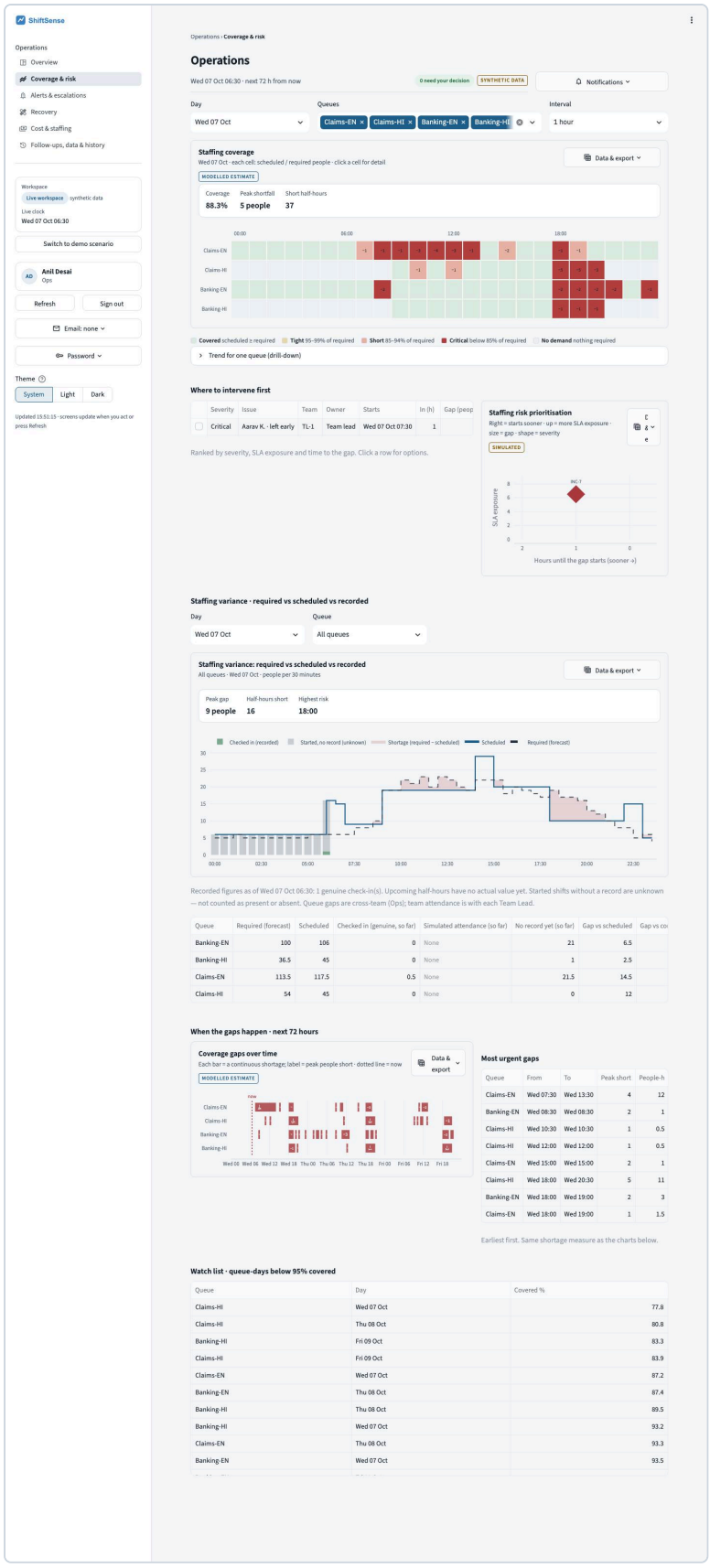
5.2 Team Lead

- **Overview:** what needs a decision first.
- **Attendance:** an agenda in time order; a started shift without a record shows *No attendance record* (unknown, never assumed absent).
- **Team schedule:** week calendar and day time-grid with a "now" line and gap rows.
- **Requests:** pending requests with the coverage effect, plus a 14-day team leave calendar. Sick-leave details stay with HR.
- **Alerts & decisions:** each staffing alert shows the gap, the proposed fix, coverage now vs after the fix, and **Approve / Modify / Reject / Escalate**. A reason is required for anything other than approve.



5.3 Ops

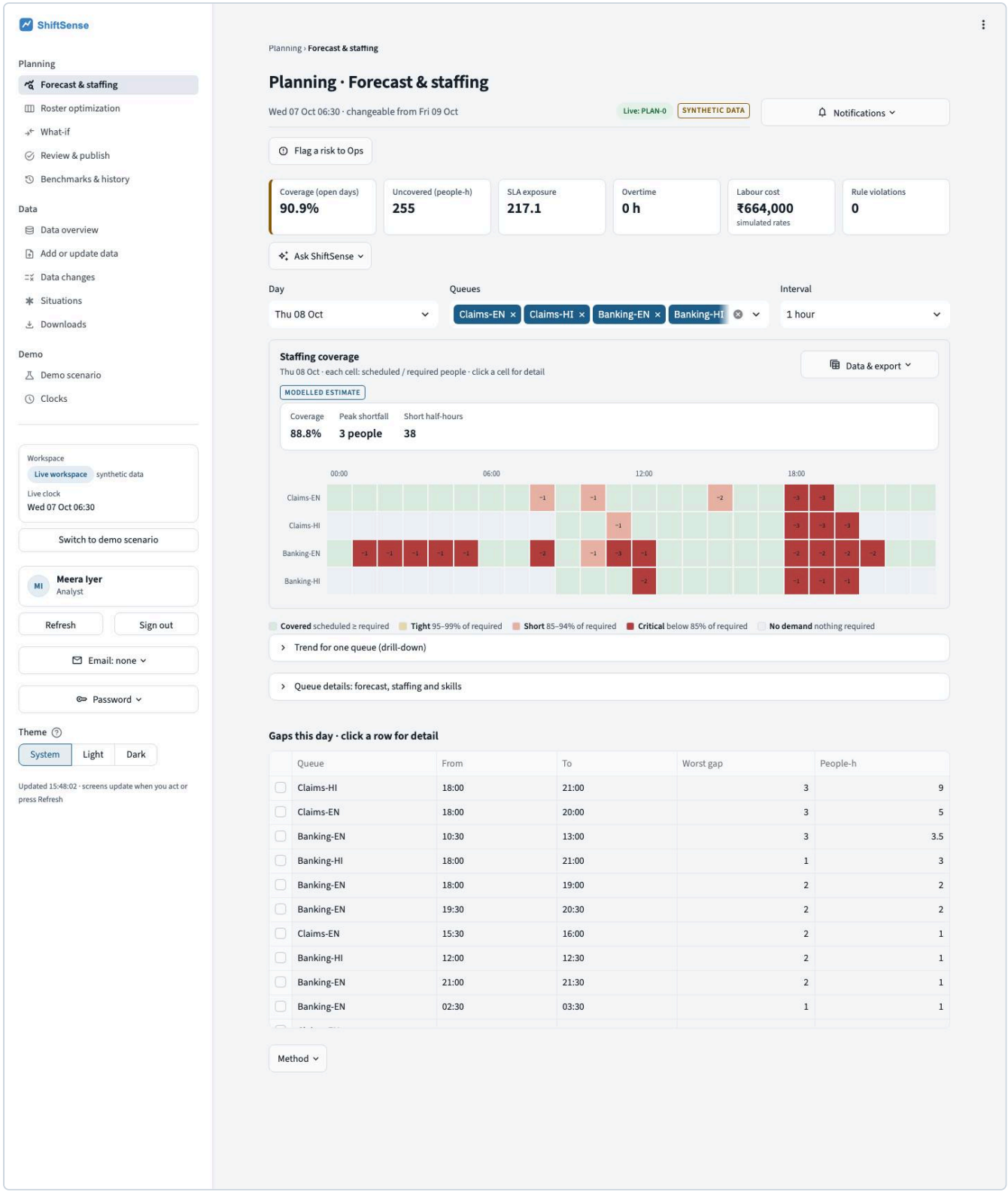
- **Overview:** exception-first control tower: open alerts, escalations, carried risk, overdue items.
- **Coverage & risk:** staffing coverage matrix (queues × time) for the next 72 h, a gap timeline with the most urgent gaps, where to intervene first, and a watch list of queue-days below 95 %.
- **Alerts & escalations:** decide escalations (overtime budget, agency cover, or accept and carry the risk, with a reason), or return them to the team lead.
- **Recovery** and **Cost & staffing:** recovery performance and simulated daily cost (labour + overtime).



5.4 Analyst

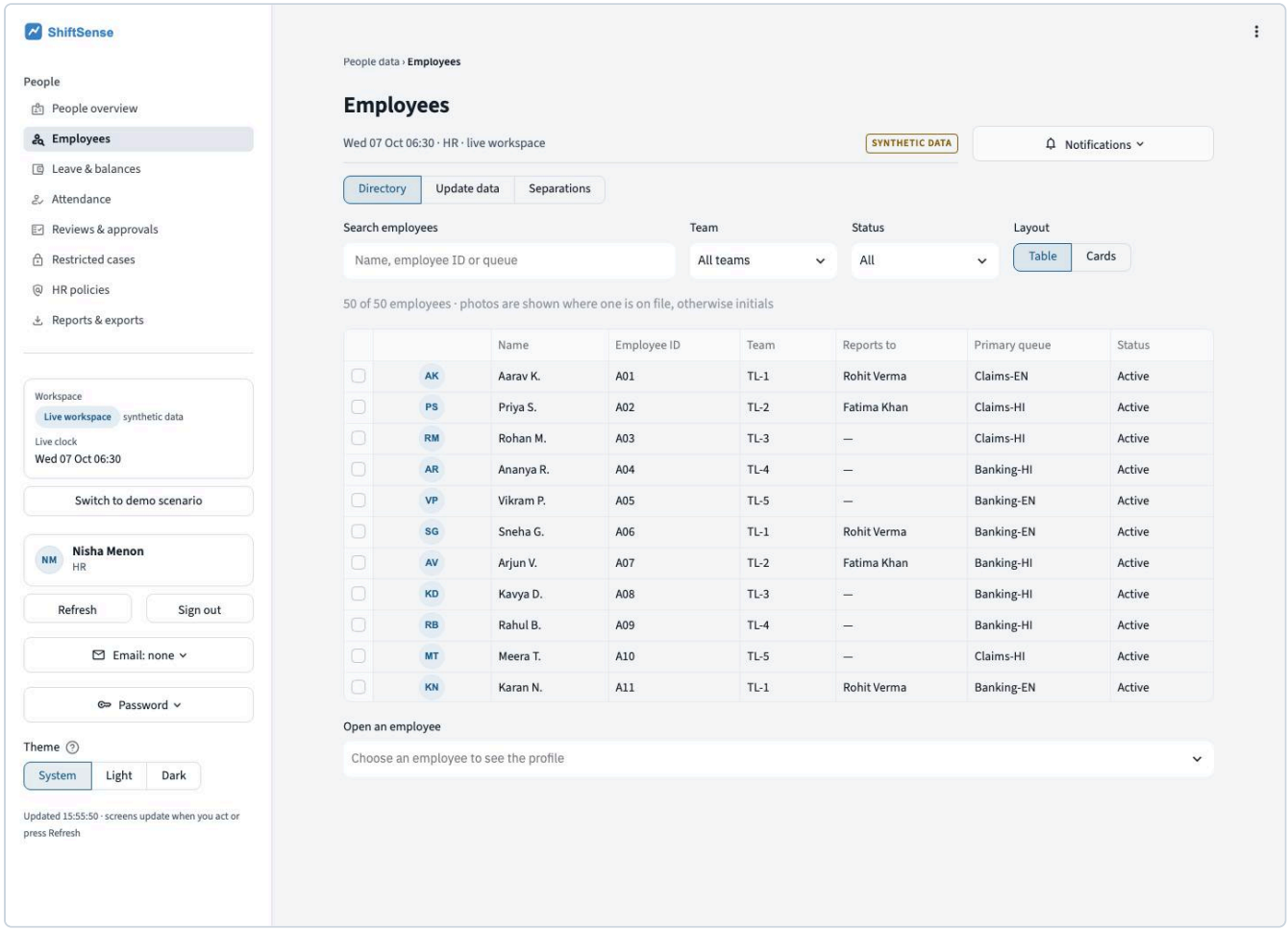
- **Forecast & staffing:** the coverage matrix (Covered / Tight / Short / Critical), filters and an interval detail card with the people who could help.
- **Roster optimization:** build a new draft with the CP-SAT optimiser, compare with the live roster and see the gaps.

- **What-if:** try scenarios without touching live.
- **Review & publish:** validate (independent rule check) → confirm → publish; restore an earlier version to roll back.
- **Data:** add or update service levels, rules, volumes and situations; follow data changes from preview to applied.
- **Demo:** load the demo story into a separate scenario and run its simulation clock. **The demo never changes live data.**



5.5 HR

- **Employees:** directory (table or cards, with photos or initials), profile with tabs, update data, separations.
- **Leave & balances:** leave calendar, opening balances (verified by a second HR user), ledger and adjustments.
- **Attendance:** correction requests after the 48-hour lock.
- **Reviews & approvals:** leave review tasks (long leave, sick leave, balance problems) and other HR approvals.
- **Restricted cases:** visible only to the assigned caseworker and Senior HR.
- **HR policies:** every HR setting with status and versions, and the retention register.



5.6 Admin

- **Overview:** the week's roster and records as a calendar or table.
- **Accounts & access:** create accounts, reset passwords, unlock, deactivate, grant Senior HR (reason required).
- **Governance & decisions:** discrepancy scans and governance cases; the decision log.
- **Audit & integrity:** the hash-chained audit trail, schedule change history and integrity checks.
- **System status:** health, backups and the email relay check.

6. Light and dark themes

Choose **Theme** → **System, Light or Dark** in the sidebar. The choice is saved on your account. Charts, calendars and dialogs follow the theme. Screenshots of both themes are in `evidence/screenshots/light-mode` and `dark-mode`.

7. Good to know

- **Live vs demo:** the demo scenario is a separate copy, marked by a banner on every screen.
- **Truthfulness:** forecasts are labelled as predictions; unknown values say *Unavailable* or *Not configured*.
- **Reasons and audit:** every decision is recorded with who, when and why, and cannot be edited.